ADITI VYAS

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GitHub: https://github.com/vyas10102 | Portfolio: https://vyas10102.github.io/portfolio2

PROFESSIONAL SUMMARY

Software Engineer with experience developing full-stack applications, REST APIs, AI-powered solutions, and QA automation. Proficient in Java, Spring Boot, React, Python, SQL, and cloud technologies, with a strong foundation in scalable backend development and secure software engineering practices.

EDUCATION

George Washington University, School of Engineering and Applied Science — August 2023 - May 2025
Master of Science in Computer Science — Washington D.C., USA
SEAS Dean Award (Merit Scholarship for Academic Excellence)

New Horizon College of Engineering — July 2019 - June 2023
Bachelor of Engineering in Computer Science — Bengaluru, India

TECHNICAL SKILLS

- **Languages:** Java | Python | JavaScript | TypeScript | SQL | Bash | C
- **Frameworks:** SpringBoot | React.js | Node.js | REST APIs | Bootstrap | HTML5 | CSS3
- **Databases:** PostgreSQL | SQL Server | MYSQL | SQLite
- **Cloud & DevOps:** AWS (S3, IAM) | Docker | Kubernetes | Git | Linux
- **Testing:** Selenium | JUnit | Postman
- **AI/ML:** YOLO | CNN | LSTM | MediaPipe | OpenCV | NLP | scikit-learn
- **Security:** JWT Authentication | Role-Based Access Control (RBAC) | Secure REST APIs | Input Validation

EXPERIENCE

Software Engineer (Full-Time, Remote)
VinfoTech Solutions LLC | July 2026 - Present
- Developed and maintained Java-based backend services and REST APIs for enterprise applications.
- Improved application stability by detecting and addressing production issues using debugging, log analysis, and root-cause analysis.
- Worked with QA and cross-functional teams to validate new features and increase software stability.
- Participated in secure deployment, code reviews, and release testing using Git-based procedures.
- Improved overall application performance by optimizing SQL queries and backend procedures.

QA Automation Engineer (Part-Time, Remote)
WeVote USA | January 2026 - June 2026
- Implemented and ran Selenium automation scripts for functional and regression testing on a React-based voting platform.
- Used Jira to identify and document over 25 software problems through manual and automated testing.
- Collaborated with developers to validate problem patches, resulting in improved application quality before production releases.
- Participated in increasing automated test coverage and decreasing repetitious manual testing efforts.

Software Engineer and AI Integration Specialist (Part-Time, Remote)
Seek and Sight | July 2025 - June 2026
- Designed and built interactive educational applications with Godot Engine and GDScript.
- Developed reusable gameplay modules and gesture-based interaction tools for neurodiverse learners.
- Optimized rendering and gaming logic to improve responsiveness and user experience.
- Collaborated with interdisciplinary teams to implement AI-assisted learning features.

Software Engineer Intern
Rovae Incorporation | January 2023 - April 2023
- Enhanced responsive client-facing websites with HTML, CSS, JavaScript, and Bootstrap.
- Collaborated with cross-functional teams to streamline UI implementation, cutting development time by 20%.
- Improved user interface performance and usability, improving customer engagement by 15%.
- Participated in testing, debugging, and deployment to ensure that functionality was consistent across browsers and platforms.

PROJECTS

ConnectWise: A Next-Gen Social Media App
GitHub | *Tech: Java, Spring Boot, React.js, PostgreSQL, AWS S3, JWT*
- Created a scalable full-stack social media platform that includes secure JWT authentication and role-based access control.
- Created REST APIs, real-time messaging, media storage using AWS S3, and PostgreSQL-backed applications using secure API design principles.

Invisibility Cloak
GitHub | *Tech: Python, OpenCV, NumPy*
- To replicate an invisibility cloak, we created a real-time computer vision program that uses HSV color segmentation and backdrop subtraction.
- Morphological procedures, temporal mask smoothing, and customizable HSV calibration for changing illumination situations all help to improve detection accuracy.

Hand Gesture Navigation System
GitHub | *Tech: Python, OpenCV, CNN, Deep Learning*
- Developed a real-time hand gesture detection system with over 90% gesture categorization accuracy for browser and operating system navigation.
- CNN-based gesture detection was developed to support touch-free scrolling, media controls, and user interface interaction.

Flight Delay Prediction
GitHub | *Tech: Python, Scikit-learn, Linear Regression, Random Forest, Gradient Boosting*
- Preprocessing airline datasets and creating predictive characteristics resulted in the development of a machine learning pipeline for predicting flight delays.
- Linear Regression, Random Forest, and Gradient Boosting models were evaluated to determine which strategy performed best in terms of prediction accuracy.

PUBLICATION

"**Space Robotics – Guardians of the Galaxy**" — IJSRSET (Vol. 9, Issue 11, 2022)
Presented at 4th National Conference on Advancements in Computer Science and Engineering.

CERTIFICATION

- AWS Cloud (S3, IAM fundamentals, Cloud Architecting)
- Red Hat DO180: Containers & Kubernetes
- Cisco CCNA v7 (Networking, Security & Automation)
- Cisco CCNA v7 (Introduction to Networks)
- NPTEL: Cloud Computing